Prabhavit Gupta

Fullstack AI Engineer | Generative AI, Enterprise RAG, Text2SQL & Production Agents

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Executive Summary

- **Production Generative AI & Client-Facing LLM Solutions:** Applied AI software engineer with 3 years commercial experience delivering client-facing AI engineering, enterprise RAG architectures, and agentic workflows. Proven track record collaborating closely with client technical stakeholders to productionize LLM solutions over unstructured documents and structured corporate data repositories.
- **Enterprise Retrieval, Text2SQL & Tool-Calling Agents:** Hands-on expertise architecting high-throughput Retrieval-Augmented Generation (RAG) pipelines, Text2SQL natural language querying engines, and autonomous tool-calling systems using LangChain, LangGraph, and Model Context Protocol (MCP). Proficient with HuggingFace, OpenAI, Anthropic models, and vector stores (pgvector, Qdrant).
- **MLOps, Cloud Deployments & Technical Advisory:** Experienced establishing robust MLOps practices, MLflow tracking, automated evaluation benchmarks (RAGAS), and containerized deployments across AWS/Azure cloud environments. Adept at translating complex data science and AI concepts into clear architectures for both executive and engineering teams.

Technical Skills

- **Generative AI & Frameworks:** LangChain, LangGraph, HuggingFace Transformers, OpenAI API, Anthropic Claude, Text2SQL, Prompt Engineering, Structured Outputs (Pydantic, JSON Schema), Guardrails against Hallucinations.
- **RAG, Vector Stores & Retrieval:** Retrieval-Augmented Generation (RAG), Semantic Chunking, Dense Vector Embeddings, PostgreSQL pgvector, Qdrant, ChromaDB, Hybrid Search (Dense + BM25), Semantic Caching (Redis).
- **MLOps & Tooling:** MLflow, Model Context Protocol (MCP Server & Client), RAGAS Evaluation Framework, Precision/Recall Benchmarking, OpenTelemetry Distributed Tracing, Structured Logging, Docker, CI/CD (GitHub Actions).
- **Core Engineering & Languages:** Python (Advanced - AsyncIO, FastAPI, PyTest, Typing), TypeScript, SQL / Data Modeling, Pandas, Scikit-Learn, PyTorch, REST/gRPC APIs, Cloud (AWS, Azure, GCP).
- **Client Engagement & Delivery:** Cross-functional Technical Discovery, Solution Architecture, Client Enablement, Agile Delivery, Systems Documentation.

Professional Experience

AI Engineer & Solutions Consultant • *Independent / AI Product Engineering Consultancy*
2024 – Present • Venice, IT (Remote Europe)

- Architected and deployed client-facing enterprise Generative AI applications and RAG systems across European client engagements, connecting LLMs to proprietary knowledge bases and SQL databases.
- Implemented Text2SQL natural language querying engines with automated schema validation and execution safety guards, allowing non-technical business users to query enterprise data stores accurately.
- Built resilient multi-agent and tool-using systems using LangGraph and Model Context Protocol (MCP), standardizing tool invocations and state management across long-running asynchronous workflows.
- Developed automated RAG evaluation harnesses using RAGAS, benchmarking precision, recall, and context faithfulness across model updates and prompt iterations.
- Designed scalable MLOps pipelines using MLflow, Docker containers, and GitHub Actions, packaging LLM microservices with OpenTelemetry tracing and structured telemetry for production reliability.

Full-Stack AI Software Engineer • *Tech Automation Systems*
2023 – 2024 • Milan / Remote, IT

- Developed high-throughput asynchronous Python and TypeScript microservices for automated document ingestion, entity extraction, and classification from complex PDF records.
- Built vector indexing pipelines and caching infrastructure using PostgreSQL and Redis, cutting retrieval latencies and stabilizing high-volume query throughput.
- Created comprehensive unit and integration test suites using PyTest, ensuring regression-proof releases across cloud environments.
- Partnered directly with enterprise customers during discovery and onboarding workshops, translating ambiguous business requirements into robust, production-ready technical architectures.

Key Engineering Projects

- **Enterprise Text2SQL & RAG Analytics Engine (Python + LangChain + PostgreSQL):** Production solution converting natural language requests into validated SQL queries and hybrid vector searches across relational and unstructured records, equipped with schema guardrails.
- **Production Agentic Knowledge Platform (LangGraph + MCP + Qdrant):** Distributed multi-agent tool execution platform utilizing Model Context Protocol connectors to query external APIs and vector stores, featuring human-in-the-loop review gates.

Education, Languages & Legal Status

- **Bachelor of Arts in Philosophy & Economics** – Ca' Foscari University of Venice (2018 – 2022)
  Rigorous education in formal logic, algorithmic decision theory, analytical reasoning, and complex systems modeling.
- **Languages:** English (C1 Professional, fluent written/spoken), Italian (Working proficiency).
- **Work Authorization & Legal Status:** EU Long-Term Resident (Permesso di Soggiorno UE a Tempo Indeterminato, Italy) – Full legal authorization to work across the EU without sponsorship; available immediately for remote engagement.